

Team Cappin

List each project member with student number (one per line)

Considering the chaos of capstone, we consider this 12-person team to be very helpful for you to alleviate the pressure of assigning stragglers.

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Project Description

LegalScrape: This project involves the development of a series of custom software tools to scrape and parse building permit data from various municipalities in the greater Montreal area. Each municipality uses a different system and document format, requiring a city-specific parser to be developed for each target. The goal is to extract key information from these unstructured sources (HTML, PDFs, etc.) and load it into a structured database.

This structured database will then be provided to the stakeholder, Munera Intelligence, for use in their downstream data analysis and construction prospecting systems.

Novelty, Innovation, and Disruption

The primary technical challenge is the lack of standardization in public data. Building permit formats, terminology, and access methods are unique to each city. A generic, scalable solution is not feasible.

The innovation in this project is therefore not in creating a single, universal tool, but in developing a robust and repeatable **methodology** for tackling these heterogeneous data sources. This involves:

- **Source Analysis:** For each city, we will analyze the specific data source, whether it's a web portal, a set of static documents, or a searchable database.
- **Custom Parser Development:** We will write a dedicated script or application for each city, employing the necessary tools (e.g., web scraping libraries for HTML, regex for text, computer vision/OCR for scanned PDFs) to accurately extract the required data fields.
- **Data Schema Mapping:** We will define a standardized database schema and write the logic to map the varied, city-specific extracted data into this consistent format.

The project is an exercise in applied data engineering, focused on solving a series of distinct but related "messy data" problems. Market disruption is a goal of the stakeholder; our focus is on the technical challenge of data acquisition and structuring.

Risk

The project's risks are primarily technical and logistical:

- **Parser Brittleness:** Each custom parser will be tightly coupled to a specific website's layout or document format. Any changes made by a municipality will break the corresponding parser, requiring manual diagnosis and code changes.
- **Technical Feasibility:** A given municipality's data may be in a format (e.g., low-quality scanned images, complex CAPTCHAs) that is prohibitively difficult to parse reliably within the project's timeframe and our current skillset.
- **Scope Creep:** The number of fields required or the number of municipalities targeted could expand, putting pressure on the project timeline. Each new city is a significant, independent unit of work.

Competition

Search terms used: "construction project leads Canada", "building permit data Canada", "construction intelligence platform Canada", "Montreal construction data"

Number of pages examined: 30+

The three most relevant competitors are:

Competitor 1: [Databid](#)

- **What they offer:** A large-scale platform for construction project listings across North America. Their approach favors breadth over depth.
- **Our Technical Distinction:** Databid's scale implies a more generalized, and likely less accurate, data collection method. Our project's novelty is its focus on creating high-fidelity, validated datasets for specific geographic areas by building custom, non-scalable parsers. This allows for a level of data quality that is difficult to achieve at continental scale.

Competitor 2: [BuildMapper](#)

- **What they offer:** A full-stack software platform using permit data for lead generation, primarily in British Columbia.
- **Our Technical Distinction:** BuildMapper provides a closed software solution. Our project is fundamentally different: we are creating a raw, structured dataset as a deliverable. Our focus is purely on the data extraction and transformation (ETL) pipeline for a market they do not serve, providing a foundational data asset rather than a user-facing application.

Competitor 3: [CCD Canada](#)

- **What they offer:** Project intelligence gathered via a team of human researchers.
- **Our Technical Distinction:** CCD's approach is manual and focused on qualitative intelligence for a few projects. Our project is a software engineering effort to systematically and comprehensively process *all* available public permit data for a given city. We are creating a quantitative, machine-readable dataset, which is a different class of information intended for automated analysis.

Description of Customer and Company

Munera Intelligence is a technology startup that uses data analysis and AI to identify early-stage construction leads.

Link: <https://www.muneraaintelligence.ca/>

Our customer is the CEO of the company: <https://ca.linkedin.com/in/dominic-macduff-auger-ab308a126>

The company's technical systems require clean, structured input data to function effectively. The raw building permit data available from municipalities does not meet this requirement. Our project directly addresses this data acquisition and pre-processing bottleneck. The custom parsing logic we develop for each city is a critical prerequisite for their business operations and expansion into new markets.